

Technical Report: LDA for Customer Sentiment Analysis

Prepared by: Emmanuel Ejima | Course: Data Science & Machine Learning
Project: Topic Modelling of McDonald's Store Reviews Using Latent Dirichlet Allocation

1. Executive Summary

This project applies Latent Dirichlet Allocation (LDA), an unsupervised machine learning algorithm, to identify main themes in 33,396 customer reviews from McDonald's stores. LDA automatically discovers hidden topics based on word co-occurrence patterns without manual review. The analysis identified six major customer experience themes: service efficiency, food quality, order accuracy, restaurant operations, dining environment, and customer service. These insights provide evidence-based recommendations for improving customer satisfaction and operational performance.

2. Methodology

2.1 Data Exploration and Preprocessing

The dataset was explored to assess structure and quality. Text preprocessing included: converting to lowercase, removing punctuation/numbers, eliminating English stop words, tokenization, and lemmatization (e.g., serving, served → serve). These steps reduced noise and improved topic interpretability.

2.2 LDA Topic Modelling

Latent Dirichlet Allocation was implemented using Gensim. Multiple topic numbers were evaluated using coherence score analysis. The optimal number of topics ($k = 6$) provided the best balance between model quality and interpretability.

3. Results

Topic	Avg Rating	Sentiment
Wait Times & Drive-Thru Service	4.43	Highly Positive
Positive Service & Food Quality	3.79	Positive
Store Hours & Operations	3.77	Neutral-Positive
Order Accuracy Issues	3.27	Neutral
Food & Drink Freshness	2.24	Negative
Dine-In Experience & Ambience	2.20	Highly Negative

Customers praised efficient drive-thru service while expressing dissatisfaction with food freshness and dine-in environment quality.

4. Discussion

Positive reviews focused on fast drive-thru service, friendly staff, and satisfactory food quality. These operational strengths directly contribute to customer satisfaction. Negative reviews centered on cold/stale food, poor cleanliness, inadequate dining environments, and occasional staff issues. These findings highlight immediate operational improvements. LDA successfully transformed thousands of unstructured reviews into actionable insights.

5. Business Recommendations

- Improve dine-in facilities through enhanced cleanliness standards, restroom maintenance, and customer service training.

- Enhance food quality control via stricter temperature monitoring and reduced food holding times to ensure freshness.

- Replicate successful drive-thru practices across all locations using standardized workflows from high-performing stores.

- Improve order accuracy through additional verification procedures before customer handoff.

- Strengthen operational consistency by ensuring accurate store hours and adequate peak/late-night staffing.

6. Conclusion

Latent Dirichlet Allocation proves effective for extracting meaningful insights from large volumes of customer reviews. By identifying six distinct themes, this analysis provides McDonald's management with evidence-based insights into operational strengths and improvement areas. Implementing the recommended actions will enhance customer satisfaction, improve service quality, and support data-driven decision-making across store operations.

Keywords: Latent Dirichlet Allocation, Topic Modelling, Customer Sentiment Analysis, Natural Language Processing, McDonald's Review Analysis